The 4Ws Problem Canvas

Who? What? Where? Why? – Structured Problem Scoping for AI Projects

Your Name

Your School / Institute

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What Is Problem Scoping?

- First and most important stage of the AI Project Cycle
- Clearly define the problem before collecting data or building models
- Identify scope, objectives, constraints, and stakeholders
- Avoids solving the wrong problem

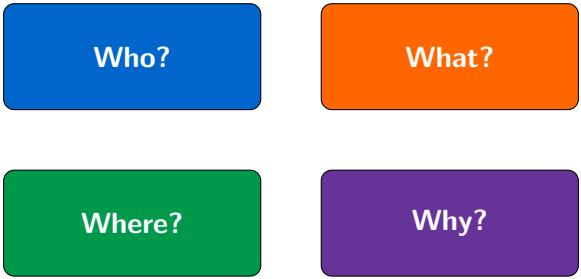


Why it matters

A well-scoped problem leads to better goals, relevant data, and useful AI solutions.

The 4Ws Problem Canvas

Problem scoping is organised around four simple questions:



These four blocks together define the problem, the people affected, the context, and the value of solving it.

Who?

Key questions

- Who are the stakeholders?
- Who is directly or indirectly affected?
- What do you know about them?

Stakeholders benefit from the solution or are involved in the problem.

- Residents, users, customers
- Government, NGOs, companies
- Experts, communities, environment



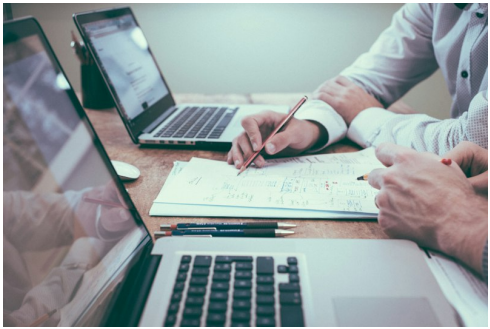
What?

Key questions

- What exactly is the problem?
- How do you know it is a problem?
- What evidence do you have?

Gather real evidence:

- Observations and reports
- News, social media, surveys
- Official statistics and data



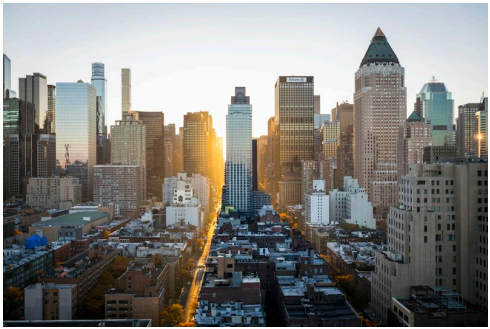
Where?

Key questions

- Where do stakeholders experience the problem?
- Physical places, online platforms, institutions?
- Is the impact local or global?

Examples of context:

- Cities, villages, schools, hospitals
- Social media, websites, apps
- Workplaces, public spaces, nature



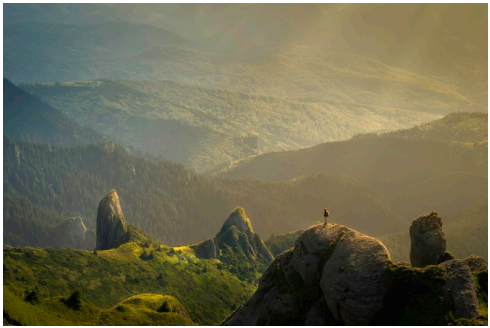
Why?

Key questions

- Why is this problem worth solving?
- How will the solution help stakeholders?
- What value does it create for society?

Typical benefits:

- Better quality of life and safety
- Efficiency and cost savings
- Sustainability and fairness



Putting the 4Ws Together

W	What it clarifies
Who?	Stakeholders and people affected
What?	The actual problem and supporting evidence
Where?	Context, location, and setting
Why?	Value and benefit of solving the problem

Result

A clear, shared understanding of the problem that guides all later stages of the AI project.

Problem Statement Template

The 4Ws are summarised into one short template that can be reused later.

Typical structure

- **Who** faces the problem
- **What** the problem is
- **Where** it occurs
- **Why** it is important to solve

Benefit

Easy to remember, easy to communicate, and useful for decision-making and project planning.

Example: Gopalapuram Air Pollution

Who: Residents, government, environmentalists, health authorities, industry

What: Rising air pollution from industry and vehicles causing health and ecological problems

Where: Residential areas, workplaces, public places, green spaces

Why: Reduce health risks, lower healthcare burden, protect biodiversity, create a sustainable city



Thank You!

Questions on the 4Ws?

